



Enhancement of calcium responses by K_{IR} current inactivation in medium spiny neurons

JOHN ERIC STEEPHEN^{1*} , MITHUN PADMAKUMAR¹ and ROHIT MANCHANDA²

¹Kerala University of Digital Sciences, Innovation and Technology,
Thiruvananthapuram 695317, India

²Indian Institute of Technology Bombay, Mumbai 400076, India

*Corresponding author (Email, johnsteephen@gmail.com)

MS received 13 February 2025; accepted 30 June 2025

Synaptic plasticity within cortico-striatal circuits critically influences learning and behavior, with the nucleus accumbens (NAc) serving as a key mediating structure. Within the NAc, medium spiny neurons (MSNs) are known to mediate input integration, whose dendritic calcium levels are thought to influence cortico-striatal plasticity. Calcium responses have been observed to correlate with firing frequency and earlier firing onset. Inward rectifying potassium (K_{IR}) currents inactivate significantly in ~40% of NAc MSNs. Studies have shown that this inactivation enhances firing frequency and advances firing onset. On the basis of these findings, we hypothesized that K_{IR} inactivation may enhance intracellular calcium levels in MSNs, with implications for synaptic plasticity. Using an 189-compartment computational model of the MSN, the influence of K_{IR} inactivation on calcium dynamics was investigated. We found that the amplitude of calcium influx was more than twice as large in the tertiary dendrite and at least 9% higher for higher input currents in response to K_{IR} inactivation. Additionally, the average calcium concentration increased by up to 26.1% in the soma, with enhancements of 4.3–21.4% in the dendrites. Our findings suggest that K_{IR} inactivation may significantly modulate synaptic plasticity, thereby impacting the learning mediated by the NAc core.

Keywords. Calcium dynamics; inactivation; inward rectifying potassium current; medium spiny neuron; nucleus accumbens; synaptic plasticity

1. Introduction

The nucleus accumbens (NAc), the major subdivision of the ventral striatum, is a structure in the basal ganglia located in the subcortical basal forebrain (Mermelstein *et al.* 1998; Salgado and Kaplitt 2015; Elibol and Şengör 2021). It is a core structure of the brain reward pathway and has been proposed as being an interface between the limbic and extrapyramidal motor systems. It is known to integrate inputs from the other structures of the reward pathway to calculate the magnitude of an expected reward and decide whether to initiate a movement to obtain it (Heimer *et al.* 1991; O'Donnell and Grace 1993; Cooper 2002; Kandel *et al.* 2021). Inhibitory GABAergic medium spiny neurons

(MSNs) make up 95% of the NAc and are its principal neurons. The rest consists of inhibitory interneurons and cholinergic cells (Pennartz *et al.* 1994; Nicola *et al.* 2000; Salgado and Kaplitt 2015). *In vivo* recordings under anesthesia have demonstrated that MSNs exhibit a two-state membrane potential regulated by active channels and synaptic input. Action potentials (APs) are fired only during depolarized up-states that occasionally disrupt the hyperpolarized down-state (Wilson 1992; Wilson and Kawaguchi 1996). Currents through one of the several ionic channels of the cell, the inward rectifying potassium (K_{IR}) channel, tend to stabilize the membrane potential in the down-state, which is close to the potassium reversal potential (E_K), by opposing excitatory inputs (Mermelstein *et al.* 1998). In the

Supplementary Information: The online version contains supplementary material available at <https://doi.org/10.1007/s12038-025-00579-3>.

physiological setting, sufficiently strong synchronous excitatory inputs from the hippocampus, prefrontal cortex (PFC), and amygdala initiate the up-state by providing the necessary synaptic depolarization, a membrane potential change that can be achieved *ex vivo* by also depolarising current injection. An MSN expresses the peptide substance P (SP) or enkephalin (ENK), or both together (ENK/SP). Inward rectifying potassium (K_{IR}) currents have been observed to inactivate substantially in ENK- and ENK/SP-positive neurons, whereas their inactivation is minimal in the case of SP-positive neurons (Mermelstein *et al.* 1998).

Intracellular calcium concentration ($[Ca^{2+}]_i$) in dendrites is believed to play an important role in synaptic plasticity (Graupner and Brunel 2012; Dorman *et al.* 2018). Cortico-striatal glutamatergic synapses have been shown to be highly plastic and their plasticity has been suggested to be dependent on MSN dendritic $[Ca^{2+}]_i$ (Jędrzejewska-Szmek *et al.* 2016; Kerr and Plenz 2002). In pyramidal neurons, back-propagating action potentials (BAPs) are known to induce changes in dendritic $[Ca^{2+}]_i$ (Hausser *et al.* 2000). Computational studies have shown the presence of BAPs in NAc MSNs (Wolf *et al.* 2005a), suggesting the existence of BAP-mediated Ca^{2+} transients in these neurons. Action potentials elicited at the soma have been shown to increase dendritic $[Ca^{2+}]_i$ in MSNs. Further, there is experimental evidence for burst strength (number of APs fired from an up-state) and time of AP onset influencing Ca^{2+} transients in striatal MSNs (Kerr and Plenz 2002, 2004). The computational study by John and Manchanda (2009) has reported the effects on dendritic $[Ca^{2+}]_i$ resulting from interactions between BAPs and excitatory post-synaptic potentials (EPSPs) in NAc MSNs. Another recent computational study (Jędrzejewska-Szmek *et al.* 2016) has suggested that the overall dynamics of calcium, and not just calcium amplitude, determines whether synaptic potentiation or depression would happen.

While it has been previously shown that K_{IR} current inactivation alters firing frequency and its onset (Steephen and Manchanda 2009), it is not known whether these translate into significant changes in calcium transients in the dendrites. In hippocampal pyramidal CA1 neurons, inactivation of K_A channels has been shown to enhance the amplitude of back-propagating action potentials and related Ca^{2+} influx (Migliore *et al.* 1999). Since dendritic $[Ca^{2+}]_i$ is believed to have a major function in the induction of synaptic plasticity, K_{IR} current inactivation, in addition to providing a higher excitability to the cell, might have an important part to play in modulating the plasticity of

the input synapses of NAc MSNs. Moderately small variations in biophysical properties have previously been shown to powerfully affect synaptic plasticity in other neuron types. For instance, in hippocampal pyramidal neurons, modulation of K_A activation kinetics has been shown to influence the induction of long-term plasticity (LTP; Watanabe *et al.* 2002). Thus, it becomes important to investigate the role of K_{IR} current inactivation on dendritic $[Ca^{2+}]_i$. The objective of this study is to compare the $[Ca^{2+}]_i$ responses of injected current and synaptic inputs to MSNs with and without K_{IR} current inactivation because of its importance for synaptic plasticity.

2. Methods

To explore the consequences of K_{IR} inactivation on dendritic $[Ca^{2+}]_i$, two previously reported model variants of the MSN (Steephen and Manchanda 2009) were employed. In one variant, there was negligible K_{IR} inactivation (non-in K_{IR} variant) and, in the other, K_{IR} inactivated significantly (in K_{IR} variant). At hyperpolarized membrane potentials, the steady-state K_{IR} current was up to 45% lower than the initial peak current in the in K_{IR} variant, whereas there was no appreciable difference between the peak and steady-state currents in the non-in K_{IR} variant. Their levels of K_{IR} inactivation matched those of the ENK/SP- and SP-positive cell types, respectively. Since a detailed description of the multi-compartment MSN model has been discussed in earlier studies (Wolf *et al.* 2005a; Moyer *et al.* 2007; Steephen and Manchanda 2009; Steephen 2011; Evans *et al.* 2013), only those features relevant to the present study are elucidated here. Each variant consisted of a soma with 4 proximal dendrites that bifurcated into 8 medial dendrites and then into 16 distal dendrites. Every distal dendrite was made up of 11 compartments, whereas the other dendrites and the soma comprised 1 compartment each, leading to a total of 189 compartments.

Although the physical spines are not modeled the additional membrane surface area attributable to the spines were included in the model by adjusting the dendritic lengths and diameters using the following equations (Segev *et al.* 1989; Wolf *et al.* 2005a, b):

$$l' = l \cdot F^{\frac{2}{3}}, \quad d' = d \cdot F^{\frac{1}{3}}, \quad F = \frac{A_{\text{dend}} + A_{\text{spines}}}{A_{\text{dend}}}$$

where l and d are the *in vitro* length and diameter, and l' and d' are the adjusted values used in the model. The

surface area parameters A_{dend} and A_{spines} were based on the published values of spine density in striatal MSP cells (Wilson 1992).

Every channel reported to be expressed in MSNs as well as AMPA, NMDA, and GABA synapses were included in the variants. Slow and fast A-type potassium (K_{As} , K_{Af}), 4-AP-resistant persistent potassium (K_{RP}), inward rectifying potassium (K_{IR}), small- and large-conductance potassium (SKKCa, BKCCa), persistent and fast sodium (NaP, NaF), high- and low-voltage-activated L-type calcium (CaL1.2, CaL1.3), as well as T-, R-, Q-, and N-type calcium (CaT, CaR, CaQ, CaN) channel currents were incorporated. The biophysical characteristics of channels and receptors as well as their distributions were obtained from previously reported studies (Wolf *et al.* 2005a; Steephen and Manchanda 2009). The locations of the channels and their peak conductances (g) or permeabilities (p) are shown in table 1. AMPA and NMDA synapses were co-localized and were distributed in the cell with one pair in each proximal dendrite, two in each secondary, and four in each tertiary dendrite adding to a total of 168 synapses. Eighty-four GABA synapses were added with 16 of them in the soma, 3 each in the proximal and medial dendrites, and 2 each in the distal dendrites. Also, 0.5% of the AMPA current and 10% of the NMDA current were designated as calcium currents (Burnashev *et al.* 1995). For every compartment, $[Ca^{2+}]_i$ in a thin shell just inside the cell membrane was

tracked by calculating calcium influx and efflux attributable to Ca-channels, active pumps, and intracellular diffusion. The intracellular diffusion coefficient used here (Jackson and Redman 2003) accounts for the combined effects of all the buffering elements. There were two Ca^{2+} pools, with N-, Q-, and R-type calcium channels (together referred to as CaNQR) contributing to one and the remaining calcium channels along with the NMDA and AMPA Ca^{2+} currents (together referred to as CaLT) contributing to the other. The net $[Ca^{2+}]_i$ for any compartment was calculated by summing the individual contributions of both the pools.

The effect of K_{IR} current inactivation on $[Ca^{2+}]_i$ for a given current input was tested by injecting current pulses of amplitude ranging from 0.25 to 0.28 nA and duration of 500 ms into the soma of both the variants. The resulting intracellular Ca^{2+} response was compared between the non- K_{IR} and K_{IR} cells. The effect of K_{IR} current inactivation on $[Ca^{2+}]_i$ for a given firing frequency was tested for the frequency range of 2–10 Hz by injecting appropriate currents into the soma of both the variants. Peak $[Ca^{2+}]_i$, defined as the peak value of the largest $[Ca^{2+}]_i$ spike, was measured from the soma, primary dendrite, secondary dendrite, and tertiary dendrite.

The effect of K_{IR} current inactivation on somatic and dendritic $[Ca^{2+}]_i$ in response to synaptic input was investigated by delivering identical synaptic input trains to both the variants and measuring the $[Ca^{2+}]_i$

Table 1. Ion channel locations and their conductance/permeability (Wolf *et al.* 2005b; Steephen and Manchanda 2009)

Channels: Na and K	g (S/cm ²)	Location
NaF	1.5	Soma
	0.0195	All dendrites
NaP	4×10^{-5}	Soma
	1.38×10^{-7}	All dendrites
K_{Af}	0.225	Soma, proximal dendrites
	0.021	Middle, distal dendrites
K_{As}	0.0104	Soma, proximal dendrites
	9.51×10^{-4}	Middle, distal dendrites
K_{IR}	1.4×10^{-4}	Soma, all dendrites
K_{RP}	0.001	Soma
BKKCa	0.001	Soma, all dendrites
SKKCa	0.145	Soma, all dendrites
Channel: Ca	p (cm/S)	Location
CaL1.2	6.7×10^{-6}	Soma, all dendrites
CaL1.3	4.25×10^{-7}	Soma, all dendrites
CaN	1×10^{-5}	Soma, all dendrites
CaQ	6×10^{-6}	Soma, all dendrites
CaR	2.6×10^{-5}	Soma, all dendrites
CaT	4×10^{-7}	Soma, all dendrites

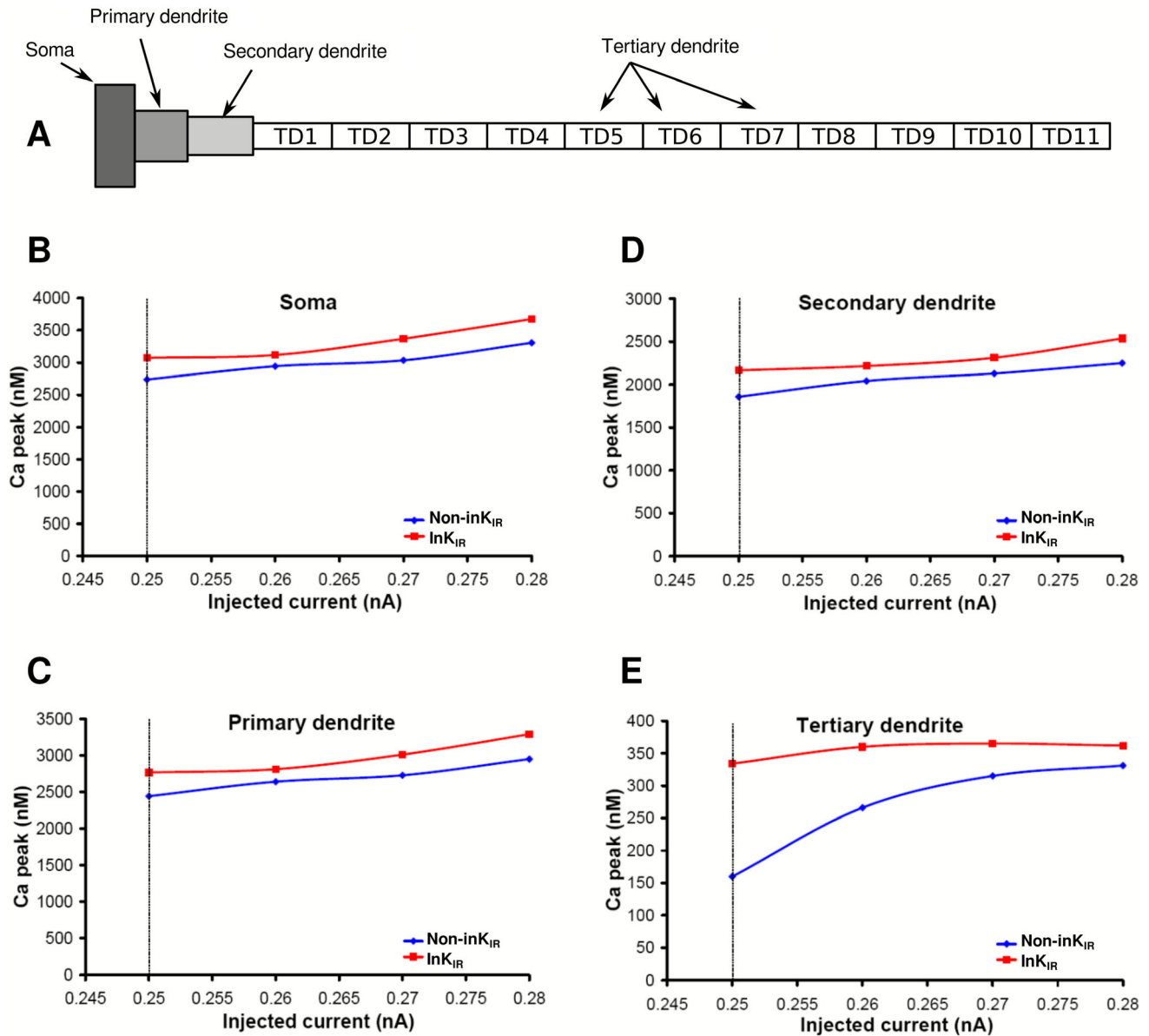


Figure 1. (A) Compartmentalization of the soma and the dendrites. The soma, primary dendrite, and secondary dendrite comprised a single compartment each, whereas the tertiary dendrite was made up of 11 compartments designated TD1 to TD11. (B–E) Ca^{2+} enhancement by K_{IR} inactivation for different current inputs in soma, and primary, secondary, and tertiary dendrites, respectively. Comparison of $[\text{Ca}^{2+}]_{\text{i}}$ peak response between non-in K_{IR} and in K_{IR} cells for a range of injected currents shows that $[\text{Ca}^{2+}]_{\text{i}}$ is enhanced by K_{IR} inactivation throughout the cell. Membrane potential and corresponding $[\text{Ca}^{2+}]_{\text{i}}$ responses of the non-in K_{IR} and in K_{IR} cells to an injected current of 0.25 nA are shown in supplementary figure 1.

response. For producing spatiotemporally distributed synaptic input, the protocol described in Steephen and Manchanda (2009) was used. The mean frequency of the Gaussian-distributed input per synapse was switched between 3 and 7.5 Hz to induce down-states and supra-threshold up-states (up-states having APs), respectively. $[\text{Ca}^{2+}]_{\text{i}}$ response was measured from the soma, primary dendrite, secondary dendrite, and all the compartments (numbering 11) of the distal dendrite.

The compartments of the distal dendrite were designated TD1 to TD11, with TD1 representing the proximal and TD11 the distal end (figure 1A). The $[\text{Ca}^{2+}]_{\text{i}}$ responses of the non-in K_{IR} and in K_{IR} cells were compared in terms of peak as well as average $[\text{Ca}^{2+}]_{\text{i}}$ during the supra-threshold up-states. Average $[\text{Ca}^{2+}]_{\text{i}}$ was estimated by calculating the total intracellular Ca^{2+} concentration during the up-state (i.e., the period when the cell was depolarized) and dividing this total

by the duration of the up-state. This provided the time-averaged calcium concentration during each up-state. Simulations were run using the NEURON simulation environment (Carnevale and Hines 2006).

2.1 Statistical analysis

Statistical analyses were conducted to assess the differences in calcium dynamics between the non- inK_{IR} and inK_{IR} model variants when synaptic input was employed. A paired two-tailed t -test was used to determine whether the differences between the two conditions were statistically significant, with a p -value less than 0.05 considered significant. The number of trials in all the cases was 30.

To quantify the magnitude of the observed differences, Cohen's d was calculated as a measure of effect size. Cohen's d was determined by dividing the difference between the means of the two groups by the pooled standard deviation. The interpretation of effect sizes followed conventional thresholds, with $d=0.2$ indicating a small effect, $d=0.5$, a medium effect, and $d=0.8$ or greater representing a large effect. All analyses were performed using custom scripts.

3. Results

3.1 Responses to injected current: identical input

Figure 1B–E shows the $[\text{Ca}^{2+}]_i$ peak responses of the non- inK_{IR} and inK_{IR} cells as a function of injected current amplitude, in the soma and the dendrites. The $[\text{Ca}^{2+}]_i$ responses along with the membrane potential changes for a particular case of 0.25 nA injected current, corresponding to the dotted lines in the figure, are shown in supplementary figure 1D. It was observed that the Ca^{2+} peak increases with injected current in both the cell types. It was also seen that irrespective of the site of measurement (primary, secondary, or tertiary dendrite or the soma) or the amount of injected current, $[\text{Ca}^{2+}]_i$ was notably enhanced by K_{IR} current inactivation. For instance, the $[\text{Ca}^{2+}]_i$ peak in the inK_{IR} cell was more than twice as large as that in the non- inK_{IR} cell in the tertiary dendrite (figure 1E; supplementary figure 1D) in response to an injected current of 0.25 nA. Although this difference decreased with higher currents, the $[\text{Ca}^{2+}]_i$ peak remained higher for the inK_{IR} cell by at least 9% (observed for 0.28 nA injected current).

3.2 Responses to injected current: identical frequency

Figure 2A–D shows the $[\text{Ca}^{2+}]_i$ peak responses of the non- inK_{IR} and inK_{IR} cells as a function of firing frequency in the soma and the dendrites. The $[\text{Ca}^{2+}]_i$ responses along with the membrane potential changes for a particular case of 8 Hz firing frequency, corresponding to the dotted lines in the figure, are shown in supplementary figure 2A–D. The $[\text{Ca}^{2+}]_i$ peak response was found to correlate with firing frequency for both the variants. Enhanced Ca^{2+} influx was present in the inK_{IR} cell even when the injected currents were matched for firing frequency in most cases. This was as high as 14% in the tertiary dendrite when the firing frequency was 8 Hz (figure 2D; supplementary figure 2D). However, the difference decreased for lower frequencies and even a reversal of the trend was observed in the tertiary dendrite for firing frequencies lower than 4 Hz (figure 2D).

3.3 Spike amplitude accommodation

Supplementary figures 1 and 2 show that in the tertiary dendrite of both cell types, Ca^{2+} influx corresponding to the first AP is the largest with subsequent APs generating smaller $[\text{Ca}^{2+}]_i$ spikes with reducing amplitude. This drop in amplitude is not very significant in the soma, the primary dendrite, and the secondary dendrite, but is very prominent in the tertiary dendrite (supplementary figures 1D and 2D). For instance, in the non- inK_{IR} cell firing at a frequency of 8 Hz, the amplitudes of the second, third, and fourth spikes reduced by only 4%, 5%, and 5%, respectively, with respect to the first spike in the soma, and this reduction was 44%, 61%, and 66% in the tertiary dendrite (supplementary figure 2A, D). This phenomenon will henceforth be referred to as 'spike amplitude accommodation' (SAA). In the tertiary dendrite, it may be observed from supplementary figures 1D and 2D that there is no correlation between $[\text{Ca}^{2+}]_i$ spike amplitudes and BAP amplitudes, suggesting that SAA is not membrane potential-dependent. Ca^{2+} currents of the non- inK_{IR} cell were examined to investigate the reasons for SAA and to determine why it is prominent in the tertiary dendrite but minimal in the soma and the other dendrites.

Supplementary figure 3A–F shows the different calcium currents measured from the tertiary dendrite corresponding to a firing frequency of 8 Hz. As could be predicted, the total Ca^{2+} current (designated Ca in

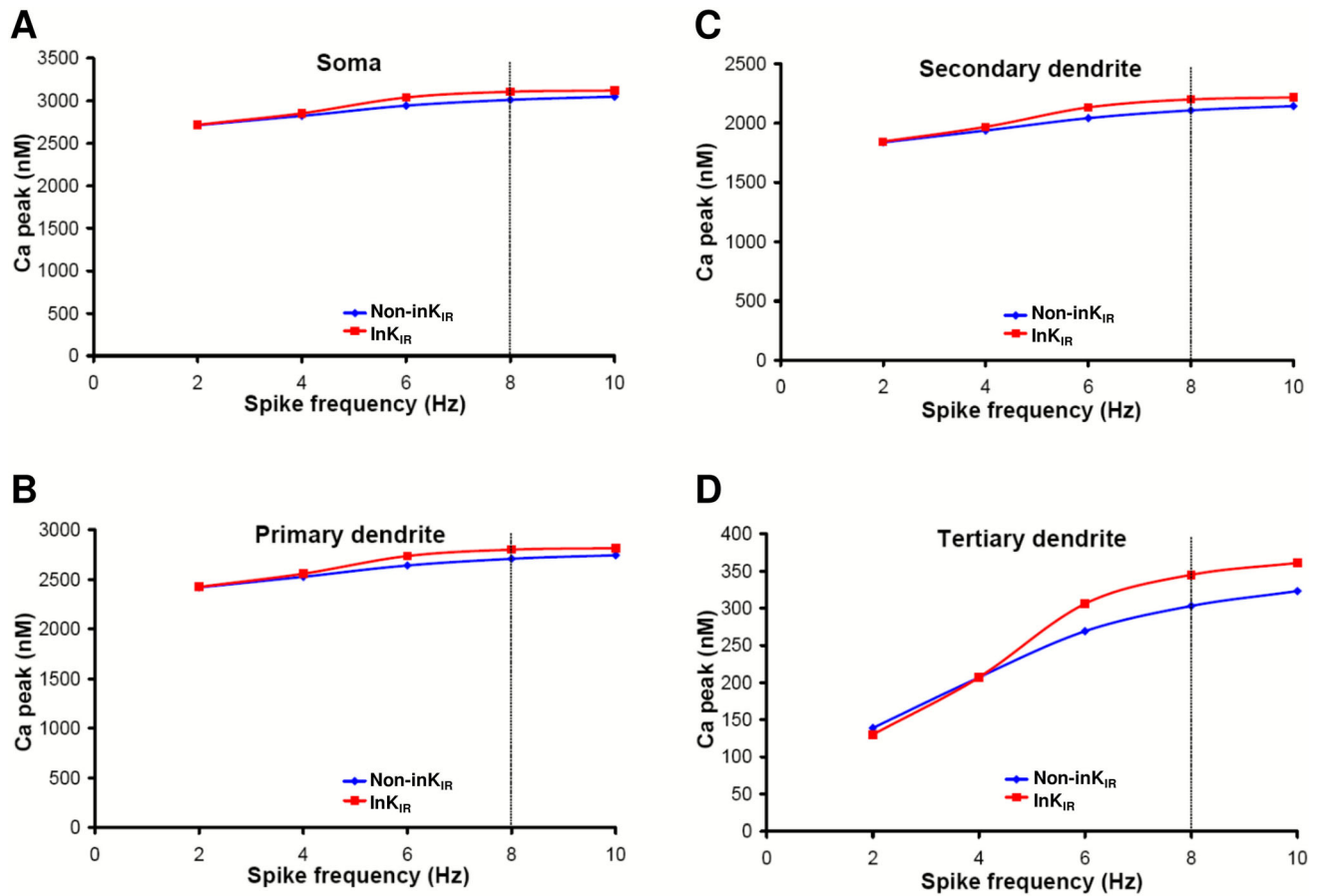


Figure 2. (A–D) Ca^{2+} enhancement by K_{IR} inactivation for different firing frequencies in soma, and primary, secondary, and tertiary dendrites, respectively. Comparison of $[\text{Ca}^{2+}]_{\text{i}}$ peak response between non-in K_{IR} and in K_{IR} cells for a range of firing frequencies showed that $[\text{Ca}^{2+}]_{\text{i}}$ is notably enhanced by K_{IR} inactivation throughout the cell in most cases.

supplementary figure 3F) correlated with the spike amplitude accommodation expressed by the $[\text{Ca}^{2+}]_{\text{i}}$ response (supplementary figure 2D). However, among the two pools of Ca^{2+} channels that contributed to the total Ca^{2+} current, this phenomenon was seen in the CaLT pool (supplementary figure 3D), but not in the CaNQR pool (supplementary figure 3C), suggesting that one or more currents of the former, comprising of the channels CaL1.2, CaL1.3, and CaT, are responsible for it. Among these three channel currents, only the CaT current underwent SAA as is evident from supplementary figure 3A, B, and E. Thus, it may be deduced that the CaT current is responsible for the SAA of the $[\text{Ca}^{2+}]_{\text{i}}$ response in the tertiary dendrite. This behaviour of the CaT current could be explained by its prolonged inactivation characteristics with an inactivation time constant in the range 37 to 127 ms.

Supplementary figure 3G shows that although the CaT current in the soma also undergoes SAA, its contribution to the total Ca^{2+} current (supplementary figure 3H) is relatively less, being 2 orders of

magnitude lower than that SAA of $[\text{Ca}^{2+}]_{\text{i}}$ is minimal in the soma. The nominal SAA in the primary and secondary dendrites are also likely to be due to the same cause.

3.4 Responses to synaptic input

3.4.1 Soma, primary dendrites, and secondary dendrites: Figure 3 shows representative membrane potential traces for two supra-threshold up-states (up-states having APs) in the non-in K_{IR} and in K_{IR} cells and the corresponding $[\text{Ca}^{2+}]_{\text{i}}$ responses at the soma, primary dendrite, and secondary dendrite. In all these sections, the inactivation of K_{IR} caused a slightly greater response in terms of peak $[\text{Ca}^{2+}]_{\text{i}}$ and notably greater response in terms of average $[\text{Ca}^{2+}]_{\text{i}}$.

In the soma, the peak $[\text{Ca}^{2+}]_{\text{i}}$ due to K_{IR} inactivation was 3.9% higher than in the absence of this inactivation (figure 4A and B), the difference being statistically significant ($t(29)=3.14$, $p<0.01$, Cohen's $d=0.42$). For

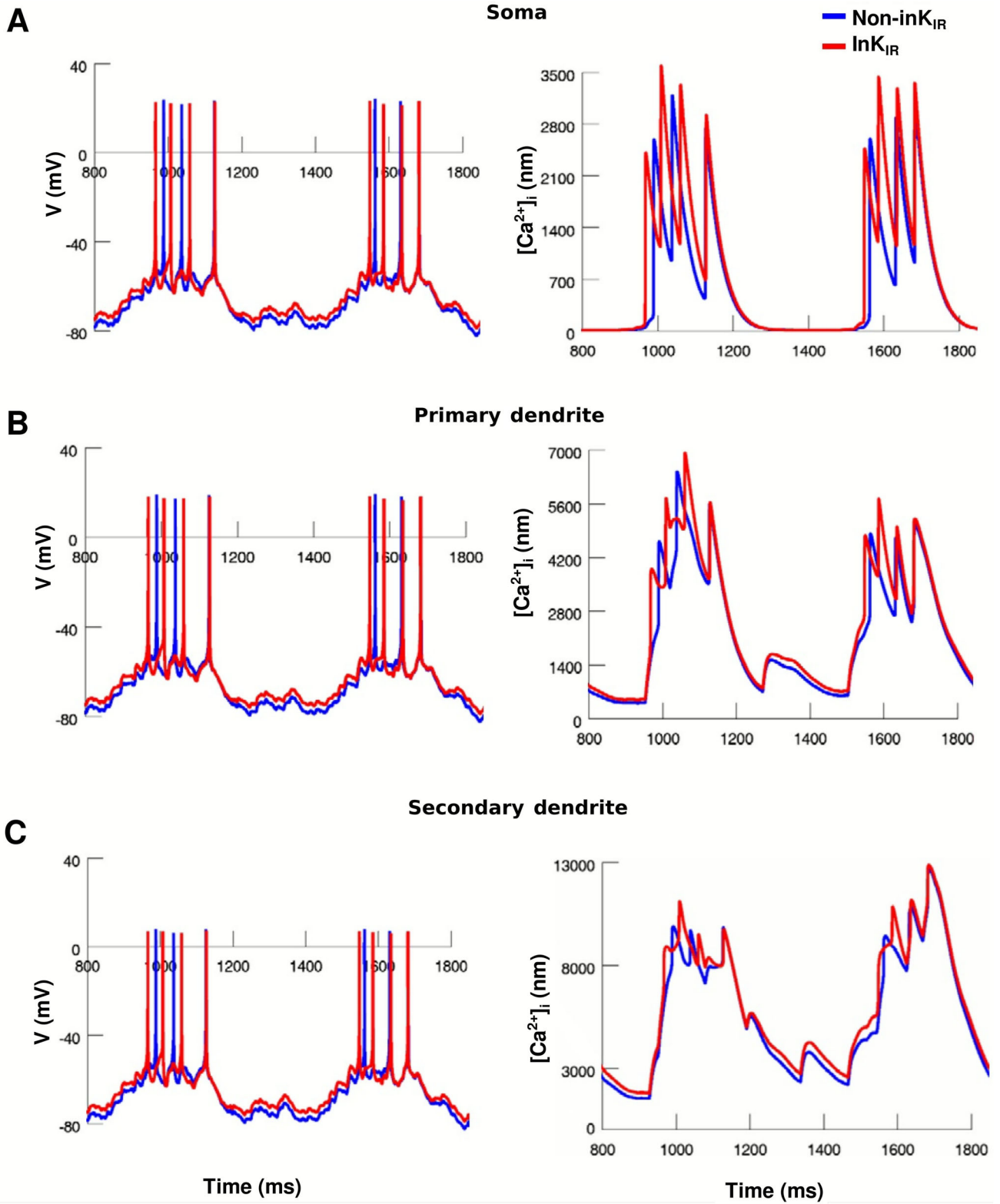


Figure 3. Representative supra-threshold up-states and corresponding $[Ca^{2+}]_i$ responses of the non-in K_{IR} and in K_{IR} cells to spatiotemporally varying synaptic input inducing up- and down-states. The in K_{IR} cell has a larger $[Ca^{2+}]_i$ response in the soma (A), primary dendrite (B), and secondary dendrite (C).

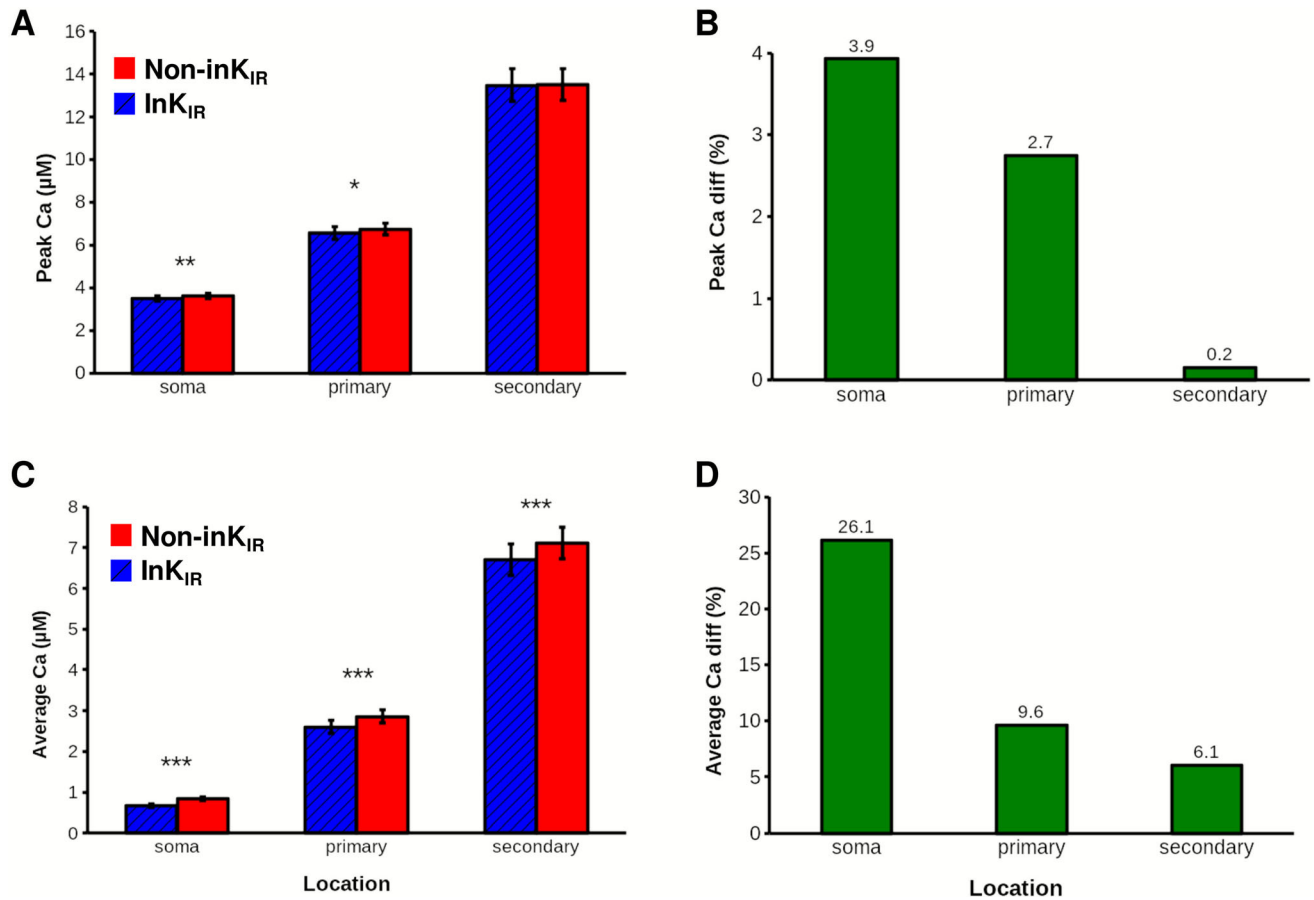


Figure 4. (A) Bar graphs (with 95% confidence interval) showing the peak intracellular Ca^{2+} concentration in the soma, primary dendrite, and secondary dendrite during supra-threshold up-states in the non-inK_{IR} and inK_{IR} cells ($n=30$). (B) Difference in peak intracellular Ca^{2+} concentration between the non-inK_{IR} and inK_{IR} cells expressed as a percentage of the non-inK_{IR} cell peak $[\text{Ca}^{2+}]_i$. (C, D) Similar bar graphs showing the average $[\text{Ca}^{2+}]_i$ in the soma, primary dendrite, and the secondary dendrite. Statistical significance of the difference between the variants has been indicated using the p -value markers (*, **, and *** indicate $p<0.05$, $p<0.01$, and $p<0.001$, respectively).

the primary dendrite, too, the difference was statistically significant ($t(29)=2.39$, $p<0.05$, Cohen's $d=0.24$) with the peak in the inK_{IR} cell larger by 2.7%. In the secondary dendrite, however, the increase in the $[\text{Ca}^{2+}]_i$ peak caused by K_{IR} channel inactivation was non-significant ($t(29)=0.18$, $p=0.85$).

In terms of average $[\text{Ca}^{2+}]_i$, the inK_{IR} cell response was significantly higher by 26.1%, 9.6%, and 6.1% for the soma ($t(29)=10.51$, $p<0.001$, Cohen's $d=1.34$), primary dendrite ($t(29)=19.34$, $p<0.001$, Cohen's $d=0.58$), and secondary dendrite ($t(29)=27.12$, $p<0.001$, Cohen's $d=0.39$), respectively (figure 4C and D). Thus, the difference between the non-inK_{IR} and inK_{IR} cells in $[\text{Ca}^{2+}]_i$ response in terms of average $[\text{Ca}^{2+}]_i$ was notably higher than that in terms of peak $[\text{Ca}^{2+}]_i$. It may also be noticed that the percentage difference was highest for the soma and it decreased with distance from the soma for peak as well as average $[\text{Ca}^{2+}]_i$.

3.4.2 Tertiary dendrite: Figure 5 shows representative membrane potential traces for two supra-threshold up-states in the two variants and the corresponding $[\text{Ca}^{2+}]_i$ responses at three locations in the tertiary dendrite.

The peak $[\text{Ca}^{2+}]_i$ in the inK_{IR} cell was 1% to 7.1% larger than that in the non-inK_{IR} cell in the various compartments of the tertiary dendrite (figure 6A and B). The difference was statistically significant in the compartments TD5 ($t(29)=2.22$, $p<0.05$, Cohen's $d=0.07$), TD7 ($t(29)=2.99$, $p<0.01$, Cohen's $d=0.10$), and TD9 ($t(29)=2.92$, $p<0.01$, Cohen's $d=0.15$), where the peak $[\text{Ca}^{2+}]_i$ response was greater in the inK_{IR} cell by 1.7%, 2%, and 2.6%, respectively. It may be noted that a statistically significant difference was observed in 3 out of the 4 compartments having glutamatergic synapses, but not in any which lacked these synapses. These were also the locations at which the Ca^{2+} influx was highest for both the variants (figure 6A).

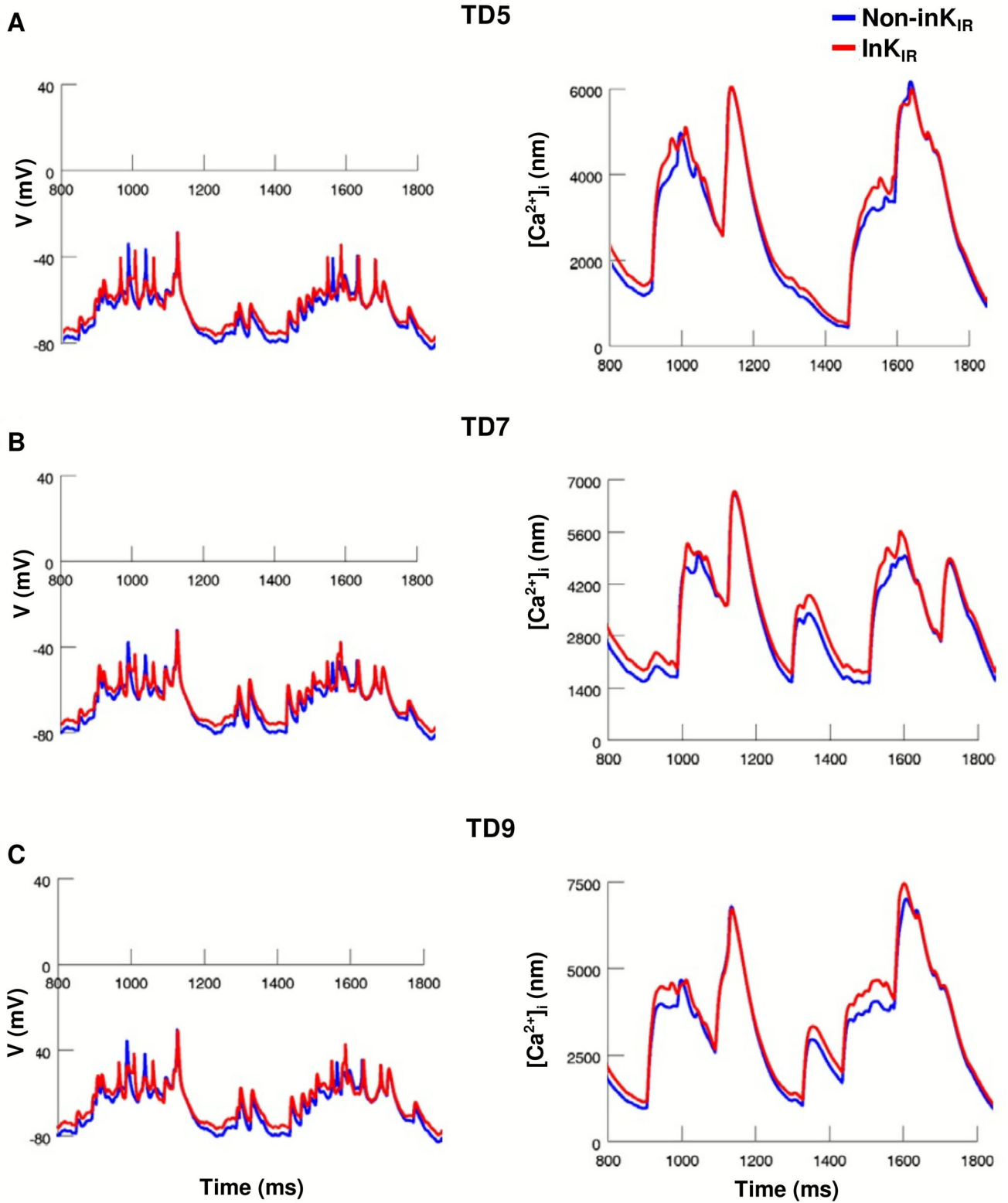


Figure 5. (A–C) Representative voltage and $[Ca^{2+}]_i$ responses at three locations in the tertiary dendrite TD (5th, 7th, and 9th compartments, respectively) corresponding to supra-threshold up-states in the soma of the non-in K_{IR} and in K_{IR} cells, induced by spatiotemporally varying synaptic input. in K_{IR} cell had a larger $[Ca^{2+}]_i$ response in terms of both peak and average $[Ca^{2+}]_i$. The corresponding responses at soma, primary dendrites, and secondary dendrites were already shown in figure 3.

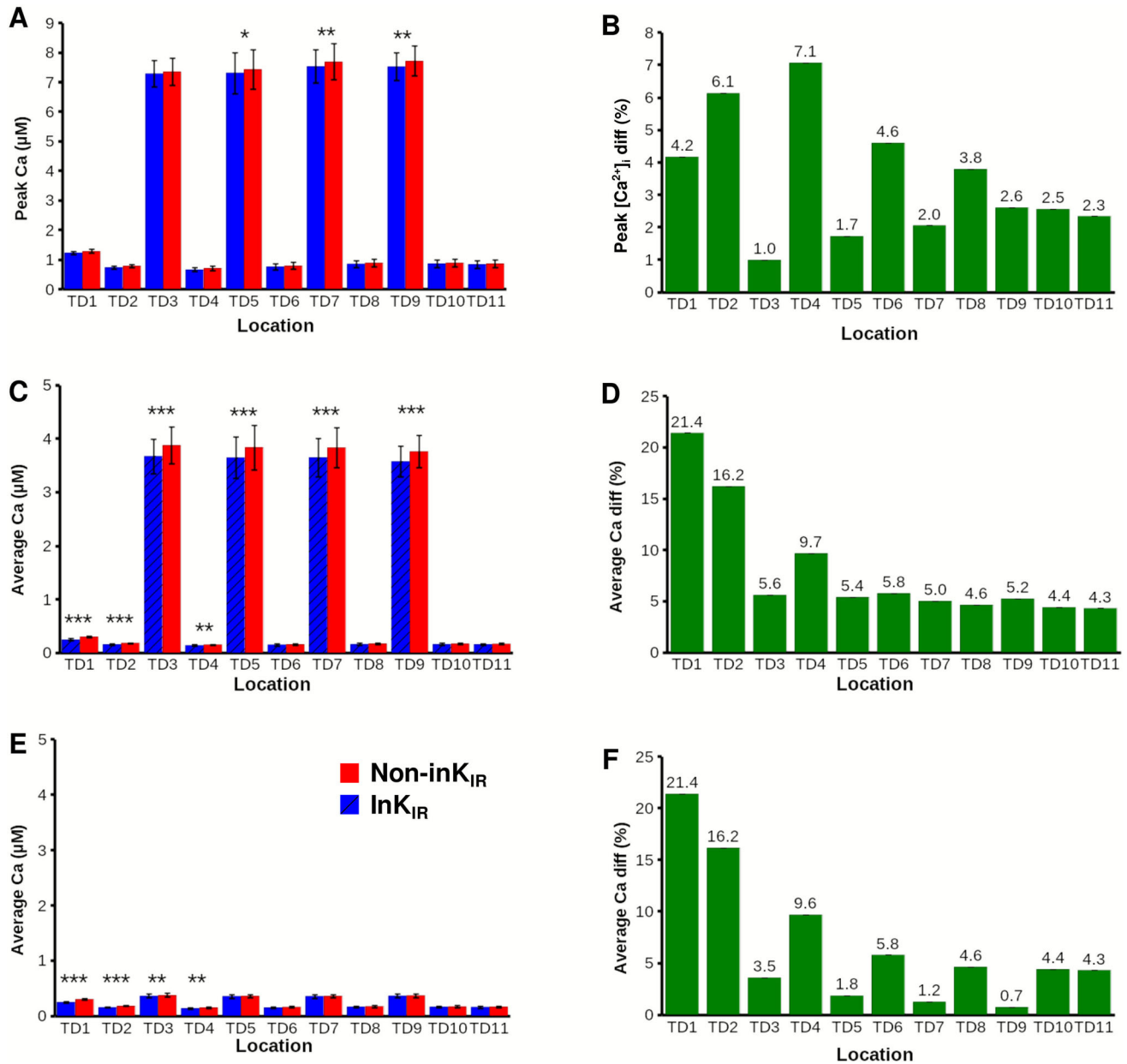


Figure 6. (A) Peak $[Ca^{2+}]_i$ in the tertiary dendrite. Bar graphs showing the peak intracellular Ca^{2+} concentration in all the compartments of the tertiary dendrite during supra-threshold up-states in the non-inK_{IR} and inK_{IR} cells ($n=30$). (B) Difference in peak intracellular Ca^{2+} concentration between the non-inK_{IR} and inK_{IR} cells expressed as a percentage of the non-inK_{IR} cell peak $[Ca^{2+}]_i$. (C, D) Average $[Ca^{2+}]_i$ in the tertiary dendrite. Bar graphs showing the average intracellular Ca^{2+} concentration (C) in all the compartments of the tertiary dendrite during supra-threshold up-states in the non-inK_{IR} and inK_{IR} cells ($n=30$). Difference in average intracellular Ca^{2+} concentration between the non-inK_{IR} and inK_{IR} cells (D) expressed as a percentage of the non-inK_{IR} cell average $[Ca^{2+}]_i$. (E, F) Average $[Ca^{2+}]_i$ in the tertiary dendrite obtained after blocking the Ca^{2+} currents through NMDA receptor channels. Statistical significance of the difference between the variants has been indicated using the p -value markers (*, **, and *** indicate $p<0.05$, $p<0.01$, and $p<0.001$, respectively).

Interestingly, a statistically significant difference was not observed for the compartments which showed the highest percentage differences (TD1, TD2, TD4, TD6, and TD8) (figure 6B). This was because, in spite of showing the highest percentage differences in the

mean, for several of the trials, the non-inK_{IR} cell had a larger $[Ca^{2+}]_i$ peak in these compartments.

In terms of average $[Ca^{2+}]_i$, the inK_{IR} cell response was 4.3% to 21.4% higher than that in the non-inK_{IR} cell (figure 6C and D). The greatest difference of

21.4% was observed in TD1, the most proximal compartment, and was statistically significant ($t(29)=8.49$, $p<0.001$, Cohen's $d=1.17$). The difference was least (4.3%) in TD11, the distal most compartment and was not statistically significant ($t(29)=1.23$, $p=0.23$). It may be noted here that these are not the compartments which had the highest and lowest peak $[Ca^{2+}]_i$ percentage differences respectively (figure 6B). Also, the percentage difference appears to demonstrate a decreasing trend as we go from the proximal to the distal end of the tertiary dendrite as can be observed from figure 6D.

The difference in average $[Ca^{2+}]_i$ was significant for all the locations having glutamatergic receptors, namely, TD3 ($t(29)=15.07$, $p<0.001$, Cohen's $d=0.23$), TD5 ($t(29)=12.01$, $p<0.001$, Cohen's $d=0.18$), TD7 ($t(29)=14.34$, $p<0.001$, Cohen's $d=0.19$), and TD9 ($t(29)=14.96$, $p<0.001$, Cohen's $d=0.24$). The locations with glutamatergic synapses were also the ones with the highest Ca^{2+} influx in both the variants. Among the compartments in which these receptors were absent, all the proximal compartments up to TD4, i.e., TD1 ($t(29)=8.49$, $p<0.001$, Cohen's $d=1.17$), TD2 ($t(29)=5.91$, $p<0.001$, Cohen's $d=0.89$), and TD4 ($t(29)=3.00$, $p<0.001$, Cohen's $d=0.45$), exhibited significant differences, whereas for all the compartments distal to it, the difference was non-significant ($p>0.05$). While only 27% of the compartments (3 out of 11) showed a significant difference in peak $[Ca^{2+}]_i$ between the non- inK_{IR} and inK_{IR} cells (figure 6A), 64% of the compartments (7 out of 11) showed significant differences in average $[Ca^{2+}]_i$ (figure 6C). Thus, the difference between the non- inK_{IR} and inK_{IR} cells in $[Ca^{2+}]_i$ response in terms of average $[Ca^{2+}]_i$ was notably higher than that in terms of peak $[Ca^{2+}]_i$. It may also be noted that all the compartments that showed significant differences in peak $[Ca^{2+}]_i$ have also shown the same in average $[Ca^{2+}]_i$.

These observations also suggested that Ca^{2+} currents via glutamatergic receptors play an important role in the difference between the non- inK_{IR} and inK_{IR} cells in average $[Ca^{2+}]_i$ in the distal compartments of the tertiary dendrite. Further, since Ca^{2+} currents comprise 10% of the NMDA current as opposed to just 0.5% of the AMPA current (see Methods), it could be speculated that NMDA receptor channel Ca^{2+} current would be primarily responsible for this. To confirm this supposition, the average $[Ca^{2+}]_i$ for both the cells (figure 6E) and the percentage differences in the average $[Ca^{2+}]_i$ between them (figure 6F) were estimated after blocking the Ca^{2+} currents through these channels. It was found that only four compartments (TD1–TD4),

all proximal, showed significant differences. TD5, TD7, and TD9 that showed significant differences in the presence of NMDA channel Ca^{2+} currents failed to do so in their absence, confirming our conjecture that NMDA receptor channel Ca^{2+} currents are essential for eliciting differences between the non- inK_{IR} and inK_{IR} cells in average $[Ca^{2+}]_i$ in the distal compartments of the tertiary dendrite.

4. Discussion

Steephen and Manchanda (2009) and Steephen (2011) previously investigated the effects of K_{IR} channel inactivation on MSN excitability. Their findings demonstrated that K_{IR} channel inactivation enhanced MSN excitability, reflected in increased firing frequency and earlier firing onset (Steephen and Manchanda 2009). Additionally, Kerr and Plenz (2002, 2004) established a correlation between $[Ca^{2+}]_i$ and both firing frequency and firing onset in MSNs. Based on these studies, we hypothesized that K_{IR} channel inactivation might also enhance $[Ca^{2+}]_i$ responses. This study aimed to investigate the impact of K_{IR} channel inactivation on $[Ca^{2+}]_i$ dynamics.

Ca^{2+} influx into the cell, resulting in an increased $[Ca^{2+}]_i$, is membrane potential-dependent and BAPs and EPSPs are the two mechanisms that cause it. Our simulations using injected current input provided a controlled environment, devoid of the complexities of temporal and spatial randomness of synaptic input and their integration, enabling us to isolate the effects of BAPs alone on $[Ca^{2+}]_i$ responses as well as to identify the parameters that influence their properties. On the other hand, our simulations using synaptic input provided us with a more physiologically realistic scenario, inclusive of the effects of BAPs and EPSPs as well as their interactions, on $[Ca^{2+}]_i$ responses. Although the focus of this investigation was on the changes in $[Ca^{2+}]_i$ responses arising from K_{IR} channel inactivation and not how properties such as firing frequency and its onset could influence $[Ca^{2+}]_i$ in general, as in studies such as those by Kerr and Plenz (2002, 2004), we have nonetheless been able to make some observations on these aspects as well.

The most important finding from this study was that irrespective of the protocol used or the location of measurement, K_{IR} inactivation boosts the responses in terms of both peak and average $[Ca^{2+}]_i$ in most cases. The simulations using injected currents showed that for a given input, K_{IR} inactivation increased the peak $[Ca^{2+}]_i$ response across the MSN soma and its

dendrites. Even when the inputs were adjusted so as to make both the variants fire with the same frequency, it was found that this inactivation enhanced the peak $[Ca^{2+}]_i$ response for most frequencies. The difference in peak $[Ca^{2+}]_i$, although not as marked as obtained with injected current input, was nonetheless significant under simulations using synaptic input too. In terms of average $[Ca^{2+}]_i$, which is a measure of sustained intracellular Ca^{2+} levels, the difference arising from K_{IR} channel inactivation was even higher. Our observation that the difference in average $[Ca^{2+}]_i$ being markedly higher than that in peak $[Ca^{2+}]_i$ assumes significance in the light of the report that in addition to the amplitude of calcium concentration, the duration of calcium elevation is crucial for synaptic plasticity (Evans and Blackwell 2015; Jędrzejewska-Szmek et al. 2016).

In MSNs, the correlation between injected current input strength and firing frequency has been already established (Steephen and Manchanda 2009). Therefore, our simulations that involved giving identical injected current input to both the cells, by showing a correlation between input strength and peak $[Ca^{2+}]_i$ response, also suggested a correlation between firing frequency and the latter. Further, this was strongly supported by our simulations in which the inputs were adjusted to cause the cell types to fire at matched frequencies. However, while such a correlation would generally be expected due to the summation of $[Ca^{2+}]_i$ spikes, as in the experimental studies of Kerr and Plenz (2002), it does not appear to occur here probably because within the range of firing frequencies explored, the time constant of the falling phase of these spikes was insufficient to elicit the effect. Further, in the presence of summation, it would be the last spike, and not the first spike as was seen in our simulations, that would have the largest peak.

Our simulations also demonstrated amplitude accommodation in $[Ca^{2+}]_i$ spikes for which the cause was traced to the inactivation of the T-type Ca^{2+} current. We believe that the correlation between firing frequency and peak $[Ca^{2+}]_i$ also arose from CaT current inactivation. The observations on SAA suggest that the farther the $[Ca^{2+}]_i$ spike from the depolarization ramp, the lower is the CaT current and smaller the spike. Since an earlier AP onset also happens as an incidental outcome of a larger firing frequency, advancing the first $[Ca^{2+}]_i$ spike also in the process, the $[Ca^{2+}]_i$ spike is able to have a larger amplitude as the inactivation of CaT is just beginning to set in. Thus, it may be argued that it is the timing of the first AP that determined the $[Ca^{2+}]_i$ peak rather than the firing

frequency. It should also be added here that this relation would be stronger in the tertiary dendrite as compared with the soma and other dendrites since the CaT current has a significantly higher contribution in determining $[Ca^{2+}]_i$ in the tertiary dendrite as was shown in section 3.3.

The earlier occurrence of the AP appears to also underpin the larger $[Ca^{2+}]_i$ peak induced by K_{IR} inactivation, as this inactivation tends to advance the AP onset in most cases. When it fails to do so, such as observed for a firing frequency of 2 Hz when the AP onset is delayed by K_{IR} inactivation (figure 5A in Steephen and Manchanda 2009), the inactivation is found to attenuate the $[Ca^{2+}]_i$ peak, especially in the tertiary dendrite (figure 2), supporting our conjecture that K_{IR} inactivation influences the $[Ca^{2+}]_i$ peak via the timing of the first AP. The relatively larger difference in peak $[Ca^{2+}]_i$ in the tertiary dendrite for a firing frequency of 2 Hz as compared with the rest of the cell may be due to the greater influence of CaT current on $[Ca^{2+}]_i$ in the tertiary dendrite. In short, K_{IR} inactivation advances the AP, causing a greater influx of CaT current, leading to a larger peak $[Ca^{2+}]_i$.

The simulations involving synaptic input differed from those using injected current input in that the $[Ca^{2+}]_i$ responses were contributed by a combination of BAPs, EPSPs, and their interactions, and not BAPs alone as was with injected current input. It was observed that in the up-state, it was not always the first AP that gave rise to the largest $[Ca^{2+}]_i$ peak unlike in the case of injected current input. Further, unlike with injected current input, instances of $[Ca^{2+}]_i$ spike summation were also seen (figures 3 and 5). Summation happened when $[Ca^{2+}]_i$ spikes were sufficiently close to each other, and was thus a function of instantaneous firing frequency rather than the average firing frequency in an up-state. The membrane potential changes resulting from EPSPs and BAPs as well as their interactions were the other factors that determined the size of $[Ca^{2+}]_i$ spikes. Thus, unlike with simulations using injected current, the largest $[Ca^{2+}]_i$ spike in an up-state was determined by factors such as summation and BAP-EPSP interaction in addition to CaT currents. It would be expected that in the soma, due to the smaller contribution of CaT currents and absence of glutamatergic synapses, $[Ca^{2+}]_i$ spike summations would be the major deciding factor of the peak $[Ca^{2+}]_i$. In primary and secondary dendrites, due to the presence of synapses, EPSPs would also play a role in addition to summation, although a relatively small one as their amplitudes would be considerably lower than those of BAPs (figure 3B and C). In the tertiary

dendrites, especially the ones with glutamatergic synapses, EPSPs and CaT currents would be the major deciding factors as BAPs would not have a major impact, these signals being severely attenuated at these locations (figure 5). The effect of the CaT currents would probably be more on the EPSPs, rather than the attenuated BAPs. The average $[Ca^{2+}]_i$, being a measure of the total Ca^{2+} influx during a supra-threshold up-state rather than a measure of peak Ca^{2+} influx, would however be dependent more on the average up-state firing frequency rather than instantaneous frequency, in addition to EPSPs and CaT currents.

Although significant enhancements in AP onset (by 29%) and firing frequency (by 28%) have been shown to arise due to K_{IR} inactivation during synaptically driven up-states (Steephen and Manchanda 2009), our present simulations showed that these did not translate into comparably higher peak Ca^{2+} influxes. This was unlike what was observed with our simulations using injected currents where large differences in peak $[Ca^{2+}]_i$ were observed. In contrast to the simulations using injected currents, where the CaT current was perhaps the only deciding factor of the $[Ca^{2+}]_i$ peak in addition to BAPs, in synaptic input-based simulations, it was determined by a complex interaction of CaT currents, $[Ca^{2+}]_i$ spike summation, BAPs, and EPSPs. Thus, while a higher average firing frequency per up-state, as would happen in the presence of K_{IR} inactivation, would drive in more Ca^{2+} during the up-state and lead to notably higher average $[Ca^{2+}]_i$ (4.3–26.1%), peak $[Ca^{2+}]_i$ would be determined by factors such as instantaneous firing frequency which may only cause slight changes (0.2–7.1%). For instance, while $[Ca^{2+}]_i$ spike summation would have almost single handedly increased the $[Ca^{2+}]_i$ peak by 3.9% in the soma upon K_{IR} inactivation, the effects of CaT and EPSPs being minimal, the average $[Ca^{2+}]_i$ got enhanced by 26.1%. Thus, the changes in average $[Ca^{2+}]_i$ resulting from K_{IR} inactivation (4.3–26.1%) seem to be more in the range of changes in firing frequency (28%) and AP onset (29%) as compared with peak $[Ca^{2+}]_i$ (0.2–7.1%).

The experimental studies of Kerr and Plenz (2002, 2004) had shown that $[Ca^{2+}]_i$ in the dendrites of striatal MSNs correlate directly with burst strength and inversely with AP onset delay. Based on their studies we had predicted that through enhanced firing frequency and reduced AP onset delay, K_{IR} inactivation may also increase $[Ca^{2+}]_i$ responses, which we have observed in the present simulations. In the study by Kerr and Plenz (2002), the larger Ca^{2+} peaks with greater firing frequency resulted from the summation of

intracellular Ca^{2+} , aided by their large decay time constants (>700 ms as suggested by the measurements in their Ca^{2+} imaging studies using fluorescent dyes). The correlation they found for peak $[Ca^{2+}]_i$ with firing onset was proposed to be NMDA-dependent. However, in our studies using injected current input, the apparent correlations of $[Ca^{2+}]_i$ peak with firing frequency and onset have been shown to result from the dynamics of the CaT current. Thus, in addition to the mechanisms suggested by Kerr and Plenz (2002, 2004), we propose that CaT current dynamics may be an additional means by which higher firing frequencies and earlier firing onsets translate into larger $[Ca^{2+}]_i$ peaks. While our results from the simulations using synaptic input showed the existence of summation and agreed with those of Kerr and Plenz (2002, 2004) in terms of the dependence of the peak $[Ca^{2+}]_i$ on spiking frequency, the relation between AP onset and peak $[Ca^{2+}]_i$ could not be compared due to the difference in the protocols used.

Our simulations also suggested that K_{IR} inactivation-induced enhancement in $[Ca^{2+}]_i$ response is strongly NMDA-dependent. While four out of six locations (67%) with glutamatergic receptors in the cell showed statistically significant differences in peak $[Ca^{2+}]_i$, just one (soma) out of eight locations (13%) without the receptors exhibited the same. Similarly, while all the locations with these receptors (100%) demonstrated statistically significant differences in average $[Ca^{2+}]_i$, only four out of eight (50%) locations without them displayed the same. Interestingly, as shown in figure 6, although the absolute change in calcium concentration caused by K_{IR} inactivation was larger in the compartments with synaptic input, the fractional difference was more pronounced in compartments without synaptic input. This suggests that the relative impact of K_{IR} inactivation on calcium dynamics is likely to be more pronounced in compartments lacking direct synaptic input. Our simulations after blocking Ca^{2+} currents via NMDA channels in the tertiary dendrite further confirmed the robust dependency of the difference in $[Ca^{2+}]_i$ response on NMDA receptors by showing that none of its distal compartments (TD5–TD11) are capable of generating significant differences in average $[Ca^{2+}]_i$ without NMDA channel $[Ca^{2+}]_i$ currents.

Our findings thus reaffirmed and strengthened our contention, which arose from the studies of Kerr and Plenz (2002, 2004) and our own past studies of MSN excitability (Steephen and Manchanda 2009), that K_{IR} inactivation would enhance intracellular Ca^{2+} levels in NAc MSNs. Intracellular calcium dynamics have been suggested to play an important role in the synaptic

plasticity of the cortico-striatal pathway (Shindou *et al.* 2011, 2019; Yagishita *et al.* 2014). Our simulations not only showed that K_{IR} inactivation modifies intracellular Ca^{2+} levels throughout the cell, with associated implications for plasticity, but also that the modulations are most significant at the locations of glutamatergic receptors, the sites for NMDA-dependent plasticity.

Analysis of the efferent connections of NAc MSNs has shown that SP-positive neurons project mainly to the ventral tegmental area (VTA) and the ENK neurons project primarily to the ventral pallidum (Mermelstein *et al.* 1998). As mentioned in the Introduction, K_{IR} inactivation in ENK MSNs is notably higher than its negligible inactivation in SP-positive MSNs. This implies enhanced calcium dynamics and greater calcium-dependent plasticity at the input synapses of the former. Therefore, we suggest that the consequent effects of K_{IR} inactivation-mediated enhanced plasticity would be felt more in the ventral pallidum and the structures that it projects to, compared with VTA.

The NAc is made up of a central core region surrounded by an outer shell region (Elibol and Şengör 2021; Salgado and Kaplitt 2015). The MSNs of the core region differ from the shell with respect to their chemical, cellular, electrophysiological, and connectivity properties (Elibol and Şengör 2021). For instance, in the core, the MSNs are medium sized and their dendrites are densely spiny, whereas in the shell, the MSNs are much smaller with fewer and significantly less spiny dendrites (Meredith *et al.* 2008). Another difference between the regions is the tendency for the ENK/SP- and ENK-positive neurons to be preferentially located in the core, and the SP-positive neurons to be preferentially located in the shell (Rogard *et al.* 1993; Mermelstein *et al.* 1998; Prensa *et al.* 2003). Since K_{IR} inactivation is substantial in ENK- and ENK/SP-positive neurons and negligible in SP-positive neurons (Mermelstein *et al.* 1998), this indicates a greater prevalence of K_{IR} inactivation in the core than the shell region of NAc.

Functionally, the MSNs in the shell neurons appear to be associated with unconditioned reward-seeking behaviors, whereas those in the core seem to have a crucial role in guiding behavior towards a specific goal, based on learning (Goto and Grace 2008; Meredith *et al.* 2008). For instance, the shell has been reported to process exploratory or spontaneous feeding and locomotor activity and the inhibition of its output causes pronounced hyperphagia (Kelley and Swanson 1997; Stratford and Kelley 1997). In contrast, the core has been observed to facilitate operant lever-pressing by drug-associated Pavlovian stimuli (Ito *et al.* 2004) and

in the mediation of Pavlovian conditioned-approach responses to food-predicting stimuli (Parkinson *et al.* 2000). The importance of the core region over the shell region in the rate of acquisition of the lever-press response for food and the overall response rate has also been shown through lesion studies (Corbit *et al.* 2001). It has further been demonstrated that the core facilitates instrumental learning and performance by bridging the delay between action and the following reinforcing stimulus (Cardinal and Cheung 2005). To mediate such learning-associated behaviors, core MSNs would have to rely on the integration of information and enhanced synaptic plasticity at the spines, requiring a greater need for mechanisms that facilitate synaptic plasticity in the core region than in the shell (Meredith *et al.* 2008).

Synaptic plasticity has been observed to be differentially expressed in the shell vs. core MSNs, with core neurons being better equipped. As mentioned earlier, they have more dendrites and are more densely endowed with spines than shell neurons. The synaptic machinery in the core neurons also appears to be better tailored for NMDA-dependent plasticity as evidenced by studies that blocked NMDA receptors in the core and shell regions (Meredith *et al.* 2008). The present investigation indicates that the MSNs in the core, where K_{IR} inactivation is more prevalent, have calcium dynamics characteristics more conducive to the induction of synaptic plasticity than those in the shell. On the basis of this, we suggest that the greater incidence of K_{IR} inactivation in the core region could be yet another mechanism that facilitates the higher synaptic plasticity in this region that in turn aids the acquisition of new instrumental learning and conditioned responses. The NMDA-dependence of the K_{IR} inactivation-induced calcium response enhancement reported in the present study further strengthens this conjecture.

The core has been suggested to receive major afferent drives from the prefrontal cortex and dorsal hippocampus with the two inputs converging on single MSNs. Similarly, the shell has inputs from the amygdala and ventral hippocampus converging on the same MSNs (O'Donnell and Grace 1993; French and Totterdell 2002, 2003; Peleg-Raibstein and Feldon 2006). In terms of efferent connections, it has been reported that the core projects to the motor areas of the cortex via the ventral pallidum, whereas the shell region outputs strongly to VTA (Heimer *et al.* 1991; Zahm 1999). These connections, in conjunction with the differential expression of calcium-dependent plasticity in the core vs. shell neurons, would signify a greater

incidence of plasticity induction in the cortico- and dorsal hippocampus–accumbens input pathways, compared with the amygdaloid and ventral hippocampal–accumbens pathways, with the consequences of the former felt in the motor areas. However, the overall circuitry involving the core and shell MSNs is far more complex (Salgado and Kaplitt 2015), and there are still uncertainties and conflicting evidence associated with their input and output connections (Heimer *et al.* 1991; Zahm 1999; Voorn *et al.* 2004). The absence of explicitly modeled dendritic spines in our cell constitutes a limitation of this study, as MSNs are known to have them. Given their importance in synaptic calcium dynamics and plasticity, integrating physical spine models represents a significant avenue for future computational research, offering deeper insights. Additionally, while dopamine modulation plays a crucial role in shaping calcium signaling and synaptic plasticity, its interaction with K_{IR} inactivation and calcium dynamics has not been explored here and remains an important target for future investigation. Gaining more precise knowledge on the organization of the pathways involving these regions, and more specifically on the afferent and efferent connections of the SP-positive, ENK, and ENK/SP-positive MSNs would add greater clarity to the implications of our findings from this study in the physiological context.

5. Conclusion

Our computational investigation on two types of MSN models – one with non-inactivating K_{IR} channels and other with inactivating K_{IR} channels – revealed that the amplitude of calcium influx was more than twice as large in the tertiary dendrite and at least 9% higher for higher input currents in response to K_{IR} inactivation. Additionally, the average calcium concentration increased by up to 26.1% in the soma due to K_{IR} inactivation, with enhancements of 4.3–21.4% in the dendrites. Our findings suggest that K_{IR} inactivation may significantly modulate synaptic plasticity, and thereby learning mediated by the NAc core.

Acknowledgements

Financial support for this study from the Department of Biotechnology, New Delhi (Project no. BT/PR9599/Med/30/34/2007), and from the Industrial Research & Consultancy Centre, Indian Institute of Technology Bombay, is gratefully acknowledged.

Author contributions

JES and RM conceptualized the study. Formal analysis and investigation was done by JES and RM. Methodology was developed by JES and RM. JES handled the project administration, under RM's supervision. Visualization of the results were prepared by JES and MP. JES prepared the original draft, which was reviewed and edited by RM and MP. All the authors read and approved the final manuscript.

Declarations

Conflict of interest The authors declare no conflict of interest.

References

- Burnashev N, Zhou Z, Neher E, *et al.* 1995 Fractional calcium currents through recombinant GluR channels of the NMDA, AMPA and kainate receptor subtypes. *J. Physiol.* **485** 403–418
- Cardinal RN and Cheung THC 2005 Nucleus accumbens core lesions retard instrumental learning and performance with delayed reinforcement in the rat. *BMC Neurosci.* **6** 1–23
- Carnevale NT and Hines ML 2006 *The NEURON book* (Cambridge University Press)
- Cooper DC 2002 The significance of action potential bursting in the brain reward circuit. *Neurochem. Int.* **41** 333–340
- Corbit LH, Muir JL and Balleine BW 2001 The role of the nucleus accumbens in instrumental conditioning: Evidence of a functional dissociation between accumbens core and shell. *J. Neurosci.* **21** 3251–3260
- Dorman DB, Jędrzejewska-Szmek J and Blackwell KT 2018 Inhibition enhances spatially-specific calcium encoding of synaptic input patterns in a biologically constrained model. *eLife* **7** 1–28
- Elibol R and Şengör NS 2021 Modeling nucleus accumbens: A computational model from single cell to circuit level. *J. Comput. Neurosci.* **49** 21–35
- Evans RC and Blackwell KT 2015 Calcium: amplitude, duration, or location? *Biol. Bull.* **228** 75–83
- Evans RC, Maniar YM and Blackwell KT 2013 Dynamic modulation of spike timing-dependent calcium influx during corticostriatal upstates. *J. Neurophysiol.* **110** 1631–1645
- French SJ and Totterdell S 2002 Hippocampal and prefrontal cortical inputs monosynaptically converge with individual projection neurons of the nucleus accumbens. *J. Comp. Neurol.* **446** 151–165
- French SJ and Totterdell S 2003 Individual nucleus accumbens-projection neurons receive both basolateral amygdala and ventral subicular afferents in rats. *Neuroscience* **119** 19–31

- Goto Y and Grace AA 2008 Limbic and cortical information processing in the nucleus accumbens. *Trends Neurosci.* **31** 552–558
- Graupner M and Brunel N 2012 Calcium-based plasticity model explains sensitivity of synaptic changes to spike pattern, rate, and dendritic location. *Proc. Natl. Acad. Sci. USA* **109** 3991–3996
- Hausser M, Spruston N and Stuart GJ 2000 Diversity and dynamics of dendritic signaling. *Science* **290** 739–744
- Heimer L, Zahm DS, Churchill L, et al. 1991 Specificity in the projection patterns of accumbal core and shell in the rat. *Neuroscience* **41** 89–125
- Ito R, Robbins TW and Everitt BJ 2004 Differential control over cocaine-seeking behavior by nucleus accumbens core and shell. *Nat. Neurosci.* **7** 389–397
- Jackson MB and Redman SJ 2003 Calcium dynamics, buffering, and buffer saturation in the boutons of dentate granule-cell axons in the hilus. *J. Neurosci.* **23** 1612–1621
- Jędrzejewska-Szmek J, Damodaran S, Dorman DB, et al. 2016 Calcium dynamics predict direction of synaptic plasticity in striatal spiny projection neurons. *Eur. J. Neurosci.* **45** 1044–1056
- John J and Manchanda R 2009 A computational study on effects of pairing BAPs and EPSPs in medium spiny neurons of nucleus accumbens; in *Proceedings of the international conference on artificial intelligence and pattern recognition (AIPR 2009)*, Orlando, FL, USA pp 105–111
- Kandel ER, Koester JD, Mack SH, et al. 2021 *Principles of neural science* 6th edition (New York: McGraw Hill)
- Kelley AE and Swanson CJ 1997 Feeding induced by blockade of AMPA and kainate receptors within the ventral striatum: A microinfusion mapping study. *Behav. Brain Res.* **89** 107–113
- Kerr JND and Plenz D 2002 Dendritic calcium encodes striatal neuron output during up-states. *J. Neurosci.* **22** 1499–1512
- Kerr JND and Plenz D 2004 Action potential timing determines dendritic calcium during striatal up-states. *J. Neurosci.* **24** 877–885
- Meredith GE, Baldo BA, Andrezejewski ME, et al. 2008 The structural basis for mapping behavior onto the ventral striatum and its subdivisions. *Brain Struct. Funct.* **213** 17–27
- Mermelstein PG, Song WJ, Tkatch T, et al. 1998 Inwardly rectifying potassium (IRK) currents are correlated with IRK subunit expression in rat nucleus accumbens medium spiny neurons. *J. Neurosci.* **18** 6650–6661
- Migliore M, Hoffman DA, Magee JC, et al. 1999 Role of an A-type K⁺ conductance in the back-propagation of action potentials in the dendrites of hippocampal pyramidal neurons. *J. Comput. Neurosci.* **7** 5–15
- Moyer JT, Wolf JA and Finkel LH 2007 Effects of dopaminergic modulation on the integrative properties of the ventral striatal medium spiny neuron. *J. Neurophysiol.* **98** 3731–3748
- Nicola SM, Surmeier DJ and Malenka RC 2000 Dopaminergic modulation of neuronal excitability in the striatum and nucleus accumbens. *Annu. Rev. Neurosci.* **23** 185–215
- O'Donnell P and Grace AA 1993 Physiological and morphological properties of accumbens core and shell neurons recorded in vitro. *Synapse* **13** 135–160
- Parkinson JA, Willoughby PJ, Robbins TW, et al. 2000 Disconnection of the anterior cingulate cortex and nucleus accumbens core impairs pavlovian approach behavior: Further evidence for limbic cortical-ventral striatopallidal systems. *Behav. Neurosci.* **114** 42–63
- Peleg-Raibstein D and Feldon J 2006 Effects of dorsal and ventral hippocampal NMDA stimulation on nucleus accumbens core and shell dopamine release. *Neuropharmacology* **51** 947–957
- Pennartz CMA, Groenewegen HJ, Lopes FH, et al. 1994 The nucleus accumbens as a complex of functionally distinct neuronal ensembles: An integration of behavioural, electrophysiological and anatomical data. *Prog. Neurobiol.* **42** 719–761
- Prensa L, Richard S and Parent A 2003 Chemical anatomy of the human ventral striatum and adjacent basal forebrain structures. *J. Comp. Neurol.* **460** 345–367
- Rogard M, Caboche J, Julie JF, et al. 1993 The rat nucleus accumbens: Two levels of complexity in the distribution of glutamic acid decarboxylase (67 kDa) and preproenkephalin messenger RNA. *Neurosci. Lett.* **155** 81–86
- Salgado S and Kaplitt MG 2015 The nucleus accumbens: A comprehensive review. *Stereotact. Funct. Neurosurg.* **93** 75–93
- Segev I, Fleshman JW and Burke RE 1989 Compartmental models of complex neurons; in *Methods in neuronal modeling: From synapses to networks* (Cambridge: MIT Press) 63–96
- Shindou T, Ochi-Shindou M and Wickens JR 2011 A Ca²⁺ threshold for induction of spike-timing-dependent depression in the mouse striatum. *J. Neurosci.* **31** 13015–13022
- Shindou T, Shindou M, Watanabe S, et al. 2019 A silent eligibility trace enables dopamine-dependent synaptic plasticity for reinforcement learning in the mouse striatum. *Eur. J. Neurosci.* **49** 726–736
- Steephen JE 2011 Excitability range of medium spiny neurons widens through the combined effects of inward rectifying potassium current inactivation and dopaminergic modulation. *Neurocomputing* **74** 3884–3897
- Steephen JE and Manchanda R 2009 Differences in biophysical properties of nucleus accumbens medium spiny neurons emerging from inactivation of inward rectifying potassium currents. *J. Comput. Neurosci.* **27** 453–470

- Stratford TR and Kelley AE 1997 GABA in the nucleus accumbens shell participates in the central regulation of feeding behavior. *J. Neurosci.* **17** 4434–4440
- Voorn P, Vanderschuren LJMJ, Groenewegen HJ, *et al.* 2004 Putting a spin on the dorsal-ventral divide of the striatum. *Trends. Neurosci.* **27** 468–474
- Watanabe S, Hoffman DA, Migliore M, *et al.* 2002 Dendritic K^+ channels contribute to spike-timing dependent long-term potentiation in hippocampal pyramidal neurons. *Proc. Natl. Acad. Sci. USA* **99** 8366–8371
- Wilson CJ 1992 Dendritic morphology, inward rectification, and the functional properties of neostriatal neurons; in *Single neuron computation* (Eds.) T McKenna, J Davis and SF Zornetzer (San Diego: Academic Press) pp 141–171
- Wilson CJ and Kawaguchi Y 1996 The origins of two-state spontaneous membrane potential fluctuations of neostriatal spiny neurons. *J. Neurosci.* **16** 2397–2410
- Wolf JA, Moyer JT, Lazarewicz MT, *et al.* 2005a NMDA/AMPA ratio impacts state transitions and entrainment to oscillations in a computational model of the nucleus accumbens medium spiny projection neuron. *J. Neurosci.* **25** 9080–9095
- Wolf JA, Moyer JT, Lazarewicz MT, *et al.* 2005b NMDA/AMPA ratio impacts state transitions and entrainment to oscillations in a computational model of the nucleus accumbens medium spiny projection neuron. *J. Neurosci.* **25** 9080–9095
- Yagishita S, Hayashi-Takagi A, Ellis-Davies GCR, *et al.* 2014 A critical time window for dopamine actions on the structural plasticity of dendritic spines. *Science* **345** 1616–1620
- Zahm DS 1999 Functional-anatomical implications of the nucleus accumbens core and shell subterritories. *Ann. NY Acad. Sci.* **877** 113–128
- Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Corresponding editor: KAVITA BABU